

Insulin regular: Drug information - UpToDate

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For additional information see "Insulin regular: Patient drug information" and "Insulin regular: Pediatric drug information"

For abbreviations, symbols, and age group definitions show table

Brand Names: US

- HumuLIN R U-500 (CONCENTRATED) [DSC];
- HumuLIN R U-500 KwikPen;
- HumuLIN R [OTC];
- Myxredlin;
- NovoLIN R FlexPen ReliOn [OTC];
- NovoLIN R FlexPen [OTC];
- NovoLIN R ReliOn [OTC];
- NovoLIN R [OTC]

Brand Names: Canada

- Entuzity KwikPen;
- HumuLIN R;
- HumuLIN R KwikPen;
- Myxrealin;
- NovoLIN GE Toronto;
- NovoLIN GE Toronto Penfill

Pharmacologic Category

- Insulin, Short-Acting

Dosing: Adult

Dosage guidance:

Dosage form information: Regular insulin U-100 is **not** interchangeable with concentrated U-500 regular insulin.

Clinical considerations: Regular insulin U-100 (100 units/mL) is a short-acting insulin, whereas concentrated U-500 regular insulin has an onset similar to short-acting insulins and a duration similar to intermediate-acting insulins. Insulin requirements vary between patients; monitor glucose levels frequently and individualize dose.

Cadaveric organ recovery

Cadaveric organ recovery (hormonal resuscitation) (off-label use): IV (premixed solution or diluted U-100 product for IV infusion): Continuous infusion of 1 unit/hour (minimum dose) to maintain blood glucose of 120 to 180 mg/dL **or** 20 units as a bolus dose (after an IV bolus of dextrose 25 g) administered to the brain-dead donor who is hemodynamically unstable requiring significant vasopressor support; give concomitantly with levothyroxine or liothyronine (preferred), vasopressin, and methylprednisolone (Ref).

Calcium channel blocker or beta-blocker toxicity

Calcium channel blocker or beta-blocker toxicity (off-label use): Note: Optimal dosage regimen has not been determined; use for patients who are refractory to initial therapies (eg, atropine, calcium, vasopressors). Consultation with a clinical toxicologist or poison center is highly recommended.

Note: Correct hypokalemia prior to initiation of insulin therapy. For patients with baseline blood glucose <200 mg/dL, administer 50 mL of dextrose 50% IV prior to initiation of insulin therapy. Start a dextrose infusion when insulin therapy is initiated to maintain euglycemia; maintain normokalemia and euglycemia during insulin infusion and after withdrawal of insulin. Monitor blood glucose and electrolytes frequently, especially at the initiation of therapy. Concentrate all IV fluids to avoid fluid overload (Ref).

IV: 1 unit/kg bolus (using a 16 unit/mL solution (Ref)), followed by a continuous infusion at 0.5 to 1 unit/kg/hour (using a 16 unit/mL solution for IV infusion (Ref)) titrated to clinical response. Higher doses (eg, boluses of up to 10 units/kg and continuous infusions of >10 units/kg/hour) have been administered with good outcomes and minimal adverse effects. Once hemodynamic parameters have stabilized, gradually decrease insulin infusion (Ref).

Diabetes mellitus, type 1, treatment

Diabetes mellitus, type 1, treatment:

Note: Regular insulin must be used concomitantly with intermediate- or long-acting insulin (ie, multiple daily injections regimen) or in a continuous SUBQ infusion device. **The total daily doses (TDDs) presented below are expressed as the total units/kg/day of all insulin formulations combined.**

General insulin dosing (off label):

Initial TDD: SUBQ (U-100): ~0.4 to 0.5 units/kg/day in divided doses; conservative initial doses of 0.2 to 0.4 units/kg/day may be considered to avoid the potential for hypoglycemia; higher initial doses may be required in patients with obesity, or who are sedentary or presenting with ketoacidosis (Ref).

Usual TDD maintenance range: SUBQ (U-100): 0.4 to 1 units/kg/day in divided doses (Ref).

Division of TDD (multiple daily injections):

Basal insulin: Generally, 40% to 50% of the TDD is given as basal insulin (intermediate- or long-acting) in 1 to 2 daily injections (Ref).

Prandial insulin: The remaining portion (ie, 50% to 60%) of the TDD is then divided and administered before, at, or just after mealtimes depending on the formulation (eg, short-, rapid-, ultra-rapid acting) (Ref).

Dosage adjustment: Dosage must be titrated to achieve glucose control and avoid hypoglycemia. Adjust dose to maintain premeal and bedtime glucose in target range. Since combinations of agents are frequently used, dosage adjustment must address the individual component of the insulin regimen that most directly influences the blood glucose value in question, based on the known onset and duration of the insulin component.

Diabetes mellitus, type 2, treatment

Diabetes mellitus, type 2, treatment:

Note: Regular insulin U-100 is **not** interchangeable with concentrated U-500 regular insulin.

Regimens involving regular insulin U-100 (100 units/mL):

Note: May be used if glycemic targets are not met despite adequately titrated basal insulin (eg, fasting glucose levels at goal, basal insulin dose >0.5 units/kg/day) (Ref). For regimens containing basal and prandial insulin, consider discontinuing noninsulin agents other than metformin, glucagon-like peptide-1 receptor agonists, and sodium-glucose cotransporter-2 inhibitors (Ref).

Initial:

Note: Prandial insulin regimens are typically initiated as one daily injection administered before the largest meal; additional mealtime injections may be added based on results of glucose monitoring. In patients with HbA_{1c} <8% when prandial insulin is initiated, consider reducing the basal insulin daily dose by 4 units/day **or** by 10% (Ref).

SUBQ (U-100): 4 to 6 units **or** 10% of the **basal** insulin dose (Ref).

Dosage adjustment: Note: Individualize dosage adjustments based on patient-specific factors (eg, glucose levels, carbohydrate intake) (Ref).

For persistently elevated glucose levels: SUBQ (U-100): Consider intensification of dietary modifications and/or increasing the corresponding mealtime dose(s) by 1 to 2 units **or** by 10% to 15% every 3 days to achieve glycemic targets while avoiding hypoglycemia (Ref). **Note:** More aggressive dose increases (eg, by ≥5 units) may be required in patients already taking >20 units/dose of regular insulin (Ref).

For hypoglycemia: SUBQ (U-100): For unexplained mild to moderate hypoglycemia, consider decreasing the corresponding mealtime dose(s) by 10% to 20% (Ref); for hypoglycemia requiring assistance from another person or blood glucose <40 mg/dL, consider decreasing the corresponding mealtime dose(s) by 20% to 50% (Ref).

Conversion from other insulin regimens to concentrated U-500 (500 units/mL) regular insulin:

Note: May be considered in patients requiring a total daily insulin dose >200 units/day. Discontinue all other insulin products prior to initiation (Ref). Patient should be under the care of a clinician experienced with using concentrated U-500 regular insulin.

Initial: SUBQ (concentrated U-500): Initial dosing depends on proximity to glycemic targets. One strategy is to administer the current total daily insulin dose as concentrated U-500 regular insulin, rounded down to the nearest 5 units, in 2 divided doses (eg, 60% prior to morning meal, 40% prior to evening meal **or** 50% before morning and evening meals) or 3 divided doses (eg, 40% prior to morning meal and 30% prior to midday and evening meals) (Ref). **Note:** In patients close to glycemic targets (eg, HbA_{1c} ≤8%), consider reducing the initial daily concentrated U-500 regular insulin dose by 10% to 50% (Ref); some experts reduce the initial U-500 regular insulin dose by 20% even in patients far from glycemic targets (Ref).

Dosage adjustment: Note: Individualize dosage adjustments based on glucose levels (Ref).

For persistently elevated glucose levels: SUBQ (concentrated U-500): Increase daily dose by 10% to 20% weekly while avoiding hypoglycemia; in patients receiving 3 injections/day, consider increasing only 2 of the 3 daily doses (Ref).

For hypoglycemia: SUBQ (concentrated U-500): For unexplained mild to moderate hypoglycemia, consider decreasing daily dose by 10% to 30% (Ref); for severe hypoglycemia requiring assistance from another person or blood glucose <40 mg/dL, consider decreasing daily dose by 20% to 50% (Ref).

For a missed meal: SUBQ (concentrated U-500): Decrease dose by 50% for that meal (Ref).

Diabetes mellitus, gestational

Diabetes mellitus, gestational (off-label use):

Note: For use as an alternative to rapid-acting insulin analogs in patients with hyperglycemia despite lifestyle interventions, in combination with intermediate- or long-acting insulin. Individualize

treatment based on timing of hyperglycemia; some patients may only require prandial insulin with 1 meal each day (Ref).

Combination regimen (ie, multiple daily injections):

Note: The total daily doses (TDD) are expressed as the total units/kg/day of all insulin formulations combined.

Initial TDD: SUBQ (U-100): ~0.5 to 1 units/kg/day; insulin requirements are typically lowest in the first trimester and highest in the third trimester (Ref).

Division of TDD (multiple daily injections): The optimal dosing strategy has not been defined; an example regimen is provided:

Basal insulin: Generally, ~60% of the TDD is given as basal insulin (eg, NPH insulin) in 2 divided doses each day (Ref).

Prandial insulin: The remaining portion (ie, ~40%) of the TDD is then divided and administered before breakfast and supper (eg, ~60% prior to breakfast and ~40% prior to supper) (Ref).

Dosage adjustment:

During pregnancy: Dosage must be titrated to achieve glucose control and avoid hyperglycemia. Adjust dose frequently (eg, every 3 to 7 days) to maintain glucose levels in target range. Dosage adjustment must address the individual component of the insulin regimen which most directly influences the blood glucose value in question, based on the known onset and duration of the insulin component (Ref).

Postpartum: Following delivery, insulin requirements decrease rapidly. Patients treated for gestational diabetes mellitus may no longer require medication postpartum (Ref).

Diabetes mellitus, patients receiving enteral feedings

Diabetes mellitus, patients receiving enteral feedings: Note: TDD of insulin is divided into a basal component (intermediate- or long-acting insulin) and nutritional and correctional components (regular insulin or rapid-acting insulins).

Nutritional: SUBQ (U-100): 1 unit of regular insulin per 10 to 15 g of carbohydrate prior to each bolus feeding; in patients receiving continuous feeds, administer every 6 hours based on the amount of carbohydrate administered over each 6-hour period (Ref).

Correctional: SUBQ (U-100): Administer correctional insulin as needed prior to each feeding (for bolus feeds) or every 6 hours (for continuous feeds) (Ref). Dosing is individualized; one example of an empiric correctional dose is 1 to 2 units per 40 to 50 mg/dL above target glucose level; patients with known insulin resistance or who are receiving glucocorticoids may require higher correctional doses (eg, 4 units per 50 mg/dL above target glucose level) (Ref).

Diabetes mellitus, patients receiving parenteral feedings

Diabetes mellitus, patients receiving parenteral feedings:

IV (U-100; added to TPN solution): 1 unit of regular insulin per 10 g of carbohydrate added to TPN IV solution; adjust dose daily (Ref). One option is to increase the amount of regular insulin added to the TPN by two-thirds of the amount of the correctional insulin used on the previous day (Ref).

SUBQ (U-100): Administer correctional regular insulin every 6 hours as needed for hyperglycemia (Ref). Dosing is individualized; one example of an empiric correctional dose is 1 to 2 units per 40 to 50 mg/dL above target glucose level; patients with known insulin resistance or who are receiving glucocorticoids may require higher correctional doses (eg, 4 units per 50 mg/dL above target glucose level) (Ref).

Diabetes mellitus, patients undergoing surgery and using an insulin pump

Diabetes mellitus, patients undergoing surgery and using an insulin pump: SUBQ: For short procedures (eg, <2 hours), continue the usual pump “basal” insulin infusion rate, with or without a

temporary 20% to 40% rate reduction, on the morning of the procedure. For long and complex procedures, consider transitioning from the insulin pump to an IV regular insulin infusion perioperatively (Ref).

Diabetic ketoacidosis

Diabetic ketoacidosis (off-label use):

Note: Begin correction of fluid deficits, if present. **If serum potassium is <3.5 mEq/L on initial presentation, delay insulin administration until serum potassium reaches ≥ 3.5 mEq/L.** Address other electrolyte abnormalities, as needed, during insulin administration. If serum glucose is <250 mg/dL when IV insulin is started, initiate dextrose-containing IV fluids (Ref). An example of a dosing regimen is as follows; refer to institutional protocols (Ref).

IV (U-100):

Note: Administer continuous infusions using premixed solution or diluted U-100 product for IV infusion.

Initial: 0.1 units/kg/**hour** via IV infusion. **Note:** If there is a delay in initiating the IV infusion, may administer an IV bolus of 0.1 units/kg prior to starting the infusion (Ref).

Dosage adjustment: Increase the IV infusion rate (eg, by double) each hour if serum glucose does not decrease by ~50 to 70 mg/dL in the first hour. Once serum glucose is <250 mg/dL, may decrease IV infusion (eg, to 0.02 to 0.05 units/kg/**hour**) and administer dextrose-containing IV fluids until diabetic ketoacidosis (DKA) has resolved (Ref).

Transition from IV to SUBQ insulin: After resolution of DKA, may initiate a SUBQ insulin regimen (eg, with basal and prandial insulin); avoid ultra-long-acting basal insulins (eg, degludec, insulin glargine U-300 [300 units/mL]) in this setting due to their long time to reach maximal glycemic effect. Continue the IV insulin infusion for ~1 to 4 hours after the first SUBQ insulin dose to avoid rebound hyperglycemia or ketoacidosis (Ref).

Hyperglycemia, hospitalized patients

Hyperglycemia, hospitalized patients (off-label use):

Note: For use in patients with persistent hyperglycemia (eg, blood glucose ≥ 180 mg/dL on two occasions) with or without a history of diabetes. Individualize glycemic targets depending on clinical status; use of institution-specific protocols to achieve glycemic targets and minimize hypoglycemia is encouraged (Ref).

IV (premixed solution or diluted U-100 product for IV infusion):

Note: IV administration of regular insulin is preferred over SUBQ in critically ill patients and may be considered in some perioperative patients (Ref).

IV continuous infusion: Initial: 1 to 4 units per hour, then titrate to achieve target blood glucose (Ref).

Transition from IV to SUBQ insulin: Before discontinuation, transition stable ICU patients to a protocol-driven 'basal-bolus' insulin regimen, based on insulin infusion history and carbohydrate intake, to maintain glucose levels in target range; avoid ultra-long-acting basal insulins (eg, degludec, insulin glargine U-300 [300 units/mL]) in this setting due to their long time to reach maximal glycemic effect (Ref). **Note:** Administer initial dose of basal insulin ≥ 2 to 4 hours before discontinuing IV insulin infusion (Ref).

SUBQ:

Note: SUBQ administration of regular insulin may be considered in noncritically ill patients or in critically ill patients who have low insulin requirements and do not have conditions affecting absorption (eg, shock, edema) (Ref).

Correctional insulin:

Note: For use in addition to scheduled basal and nutritional insulin to achieve glycemic targets; prolonged use of correctional-only (ie, sliding scale) insulin regimens without basal insulin is discouraged (Ref).

SUBQ (U-100): Refer to institution-specific protocols; one example of an empiric correctional dose is 1 to 2 units per 40 to 50 mg/dL above target glucose level; dose is typically administered with meals (or bolus feeds) or every 6 hours (if NPO or receiving continuous feeds); patients with known insulin resistance or who are receiving glucocorticoids may require higher correctional doses (eg, 4 units per 50 mg/dL above target glucose level) (Ref).

Nutritional insulin:

Initial daily dosage:

Patients not receiving nutritional insulin prior to hospitalization:

Patients eating meals: **SUBQ (U-100):** 0.03 to 0.1 units/kg/**meal** administered with or just after meals (Ref). **Note:** Dose is individualized; consider doses at the lower end of this range in older patients and in those with renal impairment; consider doses at the higher end of this range in patients receiving glucocorticoids (Ref).

Patients receiving enteral feeds : **SUBQ (U-100):** 1 unit of regular insulin per 10 to 15 g of carbohydrate prior to each bolus feeding; in patients receiving continuous feeds, administer every 6 hours based on the amount of carbohydrate administered over each 6-hour period (Ref).

Patients receiving nutritional insulin prior to hospitalization: **SUBQ (U-100):** Continue the prehospitalization nutritional insulin dose; an empiric 20% to 50% dose reduction may be considered in patients with impaired renal function, poor nutritional intake, or admission glucose levels <100 mg/dL; higher doses may be required in patients receiving glucocorticoids (Ref).

Dosage adjustment: Adjust daily dose by 10% to 20% every 2 to 3 days to achieve glycemic targets. Consider reducing dosage for glucose levels <100 mg/dL to avoid hypoglycemia; in patients with glucose levels <40 mg/dL, larger dose reductions (eg, by 20% to 40%) may be needed (Ref).

Hyperkalemia, severe/emergent

Hyperkalemia, severe/emergent (off-label use): **Note:** Use in patients with hyperkalemia-associated ECG changes or other clinical signs consistent with hyperkalemia; **or** serum potassium >6.5 mEq/L; **or** serum potassium >5.5 mEq/L, plus significant kidney function impairment and either ongoing tissue breakdown (eg, rhabdomyolysis) or ongoing potassium absorption (eg, substantial GI bleeding) (Ref). Causes a temporary shift of serum potassium intracellularly. Consider use in combination with IV calcium to stabilize myocardial cell membranes and other methods of decreasing serum potassium and enhancing potassium removal/excretion as clinically indicated (Ref). **Safety:** Serum glucose monitoring, continuous cardiac monitoring, and serial ECGs are warranted (Ref). Administration of insulin for the treatment of hyperkalemia has been associated with errors (eg, therapy delays, dosing, wrong routes). Institutions should develop protocols and standard order sets to help reduce errors (Ref). Practice may vary; refer to institutional protocols.

IV (U-100): 10 units given as an IV bolus, along with separate administration of 25 to 50 g dextrose administered over 5 minutes (Ref). After initial dose of dextrose, some experts administer 10% dextrose continuous IV infusion at 50 to 100 mL/hour for ~5 hours (Ref). **Note:** Consider omitting dextrose if blood glucose is ≥ 250 mg/dL (Ref).

Patients with risk factors for hypoglycemia (eg, pretreatment serum glucose <140 mg/dL, kidney impairment, lower body weight):

IV (U-100): 5 units given as an IV bolus, along with separate administration of 25 g dextrose administered over 5 minutes (Ref). After initial dose of dextrose, some experts administer 10% dextrose continuous IV infusion at 50 to 100 mL/hour for ~5 hours (Ref).

Duration: May repeat dosing every 2 to 4 hours as needed; monitor serum potassium and blood glucose every hour for up to 6 hours after insulin has been administered (Ref).

Hyperosmolar hyperglycemic state

Hyperosmolar hyperglycemic state (HHS) (off-label use):

Note: Begin correction of fluid deficits; some experts may wait to initiate insulin until glucose has stopped declining with fluid correction alone (Ref). **If serum potassium is <3.5 mEq/L on initial presentation, delay insulin administration until serum potassium reaches ≥ 3.5 mEq/L.** Address other electrolyte abnormalities, as needed, during insulin administration. An example of a dosing regimen is as follows; refer to institutional protocols (Ref).

IV (U-100):

Note: Administer continuous infusions using premixed solution or diluted U-100 product for IV infusion.

Initial: 0.05 units/kg/**hour** via IV infusion. If significant ketonemia, acidosis, or other signs of diabetic ketoacidosis are present, initiate at 0.1 units/kg/**hour** (Ref).

Dosage adjustment: Increase the IV infusion rate (eg, by double) each hour if serum glucose does not decrease by ~50 to 70 mg/dL in the first hour. Once serum glucose is <250 mg/dL, decrease IV infusion (eg, to 0.02 to 0.04 units/kg/**hour**) and administer dextrose-containing IV fluids to maintain glucose levels 250 to 300 mg/dL until HHS has resolved (Ref).

Transition from IV to SUBQ insulin: After resolution of HHS, may initiate a SUBQ insulin regimen (eg, with basal and prandial insulin); avoid ultra-long-acting basal insulins (eg, degludec, insulin glargine U-300 [300 units/mL]) in this setting due to their long time to reach maximal glycemic effect. Continue the IV insulin infusion for ~1 to 4 hours after the first SUBQ insulin dose to avoid rebound hyperglycemia or ketoacidosis (Ref).

Dosage adjustment for concomitant therapy: Significant drug interactions exist, requiring dose/frequency adjustment or avoidance. Consult drug interactions database for more information.

Dosing: Kidney Impairment: Adult

Altered kidney function:

Calcium channel blocker or beta blocker toxicity: IV: No dosage adjustment necessary; titrate to clinical response (while maintaining euglycemia) (Ref).

Diabetic ketoacidosis, hyperosmolar hyperglycemic state, or cadaveric organ recovery (hormonal resuscitation) IV: No dosage adjustment necessary; titrate to target glucose levels (Ref).

Diabetes mellitus; Hyperglycemia (hospitalized patients):

The following recommendations based on kidney function are suggested (Ref):

Initial dosing:

eGFR >50 mL/minute/1.73 m²: IV, SUBQ: No dosage adjustment necessary.

eGFR 10 to 50 mL/minute/1.73 m²: IV, SUBQ: Administer 75% of the usual starting dose.

eGFR <10 mL/minute/1.73 m²: IV, SUBQ: Administer 50% of the usual starting dose.

Dosage adjustment: IV, SUBQ: Dose adjustments should generally follow the usual indication-specific recommendations but titrated more gradually.

Hyperkalemia, severe/emergent: IV (U-100):

5 units given as an IV bolus, along with separate administration of 25 g dextrose administered over 5 minutes (Ref). After initial dose of dextrose, some experts administer 10% dextrose continuous IV infusion at 50 to 100 mL/hour for ~5 hours (Ref).

Duration: May repeat dosing every 2 to 4 hours, as needed; monitor serum potassium and blood glucose every hour for up to 6 hours after insulin has been administered (AHA [Vanden Hoek 2010], Mount 2021, Moussavi 2019).

Dosing: Liver Impairment: Adult

There are no dosage adjustments provided in the manufacturer's labeling (has not been studied); dosage requirements may be reduced and patients may require more frequent dose adjustment and glucose monitoring.

Dosing: Older Adult

Refer to adult dosing; avoid this agent in the absence of basal or long-acting insulin (Ref).

Dosing: Pediatric

(For additional information see "Insulin regular: Pediatric drug information")

Dosage guidance:

Clinical considerations: Insulin regular is a short-acting insulin formulation. Insulin doses should be individualized based on patient needs; adjustments may be necessary with changes in physical activity, meal patterns, acute illness, or with changes in renal or hepatic function. Insulin requirements vary dramatically between patients and dictate frequent monitoring and close medical supervision. Insulin regimens vary widely by region, practice, and institution; consult institution-specific guidelines.

Diabetes mellitus, type 1

Diabetes mellitus, type 1: Infants, Children, and Adolescents: **Note:** Insulin regular is generally used concomitantly with intermediate- or long-acting insulin (ie, multiple daily injection regimen) or via a continuous SubQ insulin infusion pump. The daily doses presented are expressed as the total units/kg/day **of all insulin formulations combined.**

General insulin dosing:

Initial total daily insulin: SubQ: Initial: 0.4 to 0.5 units/kg/day in divided doses (Ref); usual range: 0.4 to 1 units/kg/day in divided doses (Ref); lower doses (0.25 units/kg/day) may be used, especially in young children, to avoid potential hypoglycemia (Ref); higher doses may be necessary for some patients (eg, obese, concomitant steroids, puberty, sedentary lifestyle, following diabetic ketoacidosis presentation) (Ref).

Usual total daily maintenance range: SubQ: Doses must be individualized; however, an estimate can be determined based on phase of diabetes and level of maturity (Ref).

Partial remission phase (Honeymoon phase): <0.5 units/kg/day.

Prepubertal children (not in partial remission):

Infants ≥ 6 months and Children ≤ 6 years: 0.4 to 0.8 units/kg/day.

Children ≥ 7 years: 0.7 to 1 units/kg/day.

Pubescent Children and Adolescents: During puberty, requirements may substantially increase to >1 unit/kg/day and, in some cases, up to 2 units/kg/day.

Division of daily insulin requirement (multiple daily injections):

Basal insulin: Generally, ~30% to 50% of the total daily insulin dose is given as basal insulin (intermediate- or long-acting) in 1 to 2 daily injections (Ref).

Prandial insulin: The remaining portion of the total daily dose is then divided and administered before or at mealtimes (depending on the formulation) as rapid-acting (eg, aspart, glulisine, lispro) or short-acting (regular). In most type 1 patients, the use of a rapid-acting insulin analog is preferred over regular insulin to reduce hypoglycemia risk (Ref).

Dose titration: Treatment and monitoring regimens must be individualized to maintain premeal and bedtime glucose in target range; titrate dose to achieve glucose control and avoid hypoglycemia. Since combinations of agents are frequently used, dosage adjustment must address the individual component of the insulin regimen which most directly influences the blood glucose value in question, based on the known onset and duration of the insulin component.

Surgical patients (Ref): **Note:** Diabetic patients should be scheduled as the first case of the day.

Minor surgeries:

Morning procedure: Omit short-acting insulin unless it is need to correct hyperglycemia and administer the usual morning dose of long-acting insulin or 50% to 70% of the usual morning dose of insulin NPH; alternatively, administer IV insulin (regular) infusion beginning at least 2 hours prior to surgery.

Afternoon procedure: If allowed to eat breakfast, administer 50% of the usual morning dose of short-acting insulin (regular) with breakfast. Begin an IV insulin (regular) infusion beginning at least 2 hours prior to surgery.

Postprocedure: Once normal oral intake is achieved, resume usual insulin regimen; monitor closely; insulin requirement may be higher due to risk of changes related to surgery (ie, postoperative stress, medication changes, inactivity).

Major surgeries:

Evening prior to surgery: Administer the usual evening and/or bedtime insulin(s); patients on continuous subcutaneous insulin infusion (CSII) may continue normal insulin basal rates overnight; if there is a concern for hypoglycemia, basal rate may be reduced by 20% at ~3 am.

Morning of surgery: Omit morning insulin (short- and long-acting) and start IV insulin (regular) infusion and IV dextrose at least 2 hours prior to surgery; patients on CSII, should discontinue CSII when IV insulin infusion is started.

Postprocedure: Once normal oral intake is achieved, resume usual insulin regimen; monitor closely; insulin requirement may be higher due to risk of changes related to surgery (ie, postoperative stress, medication changes, inactivity).

Diabetes mellitus, type 2

Diabetes mellitus, type 2: Children ≥ 10 years and Adolescents: SubQ: The goal of therapy is to achieve an HbA1c $< 7\%$ as quickly as possible using the safe titration of medications. Initial therapy in metabolically unstable patients (eg, plasma glucose ≥ 250 mg/dL, HbA1c $\geq 8.5\%$ and symptoms excluding acidosis) may include once-daily long-acting basal insulin (preferred) or intermediate-acting insulin (eg, NPH) in combination with lifestyle changes and metformin. In patients who fail to achieve glycemic goals with metformin and basal insulin, may consider initiating prandial insulin (rapid-acting insulin or regular insulin in select situations) and titrate to achieve goals. Once initial goal reached, insulin should be slowly tapered over 2 to 6 weeks by decreasing the insulin dose by 10% to 30% every few days and the patient transitioned to lowest effective insulin doses or metformin monotherapy if able (Ref). **Note:** Patients who are ketotic or present with ketoacidosis require aggressive management with SubQ or IV insulin to correct the hyperglycemia and ketosis/ketoacidosis. Once ketosis/ketoacidosis resolves, metformin should be initiated as appropriate, while continuing insulin via the SubQ route; insulin can be titrated as appropriate once goals are achieved.

Diabetic ketoacidosis

Diabetic ketoacidosis (DKA): Limited data available: Infants, Children, and Adolescents: **Note:** Severe DKA should be treated with IV regular insulin. Serum glucose is not a direct indicator of the overall metabolic abnormalities and may decrease more rapidly than correction of the metabolic abnormalities such as acid-base abnormalities, blood β -hydroxybutyrate (BOHB), and anion gap. As part of overall DKA management, dextrose should be added to IV fluids to prevent hypoglycemia, usually once serum glucose is between 250 to 300 mg/dL, but it may be required sooner if serum glucose has decreased precipitously. Generally, only dextrose 5% is necessary and is added to NS or $\frac{1}{2}$ NS; however, dextrose 10% or 12.5% may be necessary in some cases (Ref). Refer to institution-specific protocols where appropriate.

Continuous IV infusion:

Initial: 0.05 to 0.1 units/kg/hour (start insulin infusion at least 1 hour after starting fluid replacement therapy); DO NOT bolus at start of therapy; continue the rate at 0.05 to 0.1 units/kg/hour, if tolerated, until resolution of ketoacidosis (pH > 7.3 ; bicarbonate > 15 mEq/L, BOHB < 1 mmol/L, and/or closure of anion gap); serum BOHB should decrease by approximately 0.5 mmol/L/hour; adjust insulin if the expected rate of biochemical correction does not occur. **Note:** Some patients (eg, some young

children with DKA, older children with established diabetes) may have marked sensitivity to insulin requiring lower infusion rates; these lower infusion rates should only be used provided that resolution of the acidosis continues (Ref).

Transition from IV to SubQ insulin: Once ketoacidosis has resolved and oral intake is tolerated, transition to a SubQ insulin regimen. An overlap between discontinuation of IV insulin and administration of SubQ insulin is recommended to ensure adequate plasma insulin levels. A dose of basal (long-/intermediate-acting) insulin should be administered in addition to rapid-/short-acting insulin. To prevent rebound hyperglycemia, adjust timing of SubQ insulin administration prior to infusion discontinuation dependent on type of insulin used; for SubQ regular insulin: 1 to 2 hours, or for rapid-acting insulin: 15 to 30 minutes; with intermediate- or long-acting insulin, the overlap should be longer and the rate of IV insulin administration gradually decreased (eg, administer basal in the evening and discontinue the IV infusion the next morning); optimal timing for transition is around mealtime for convenience. Benefits have been seen with early administration of basal insulin during IV insulin infusion (Ref).

SubQ: **Note:** SubQ administration for DKA treatment may be considered for patients with uncomplicated DKA in whom peripheral circulation is adequate and continuous IV regular insulin administration is not possible (Ref).

Initial: 0.8 to 1 unit/kg/day in divided doses every 4 hours; titrate dose by 10% to 20% based on blood glucose concentration prior to next insulin dose.

Calcium channel blocker or beta-blocker toxicity

Calcium channel blocker or beta-blocker toxicity (high-dose insulin therapy): Limited data available (Ref):

Note: Consultation with a clinical toxicologist or poison control center is highly recommended.

Prior to initiating high-dose insulin therapy, dextrose supplementation is necessary in infants and children with serum glucose <400 mg/dL and in adolescents with serum glucose <200 mg/dL (Ref). Monitor serum glucose closely during therapy, with increased frequency of monitoring following the initial insulin bolus and during insulin infusion rate changes (Ref). Concomitant dextrose infusion or intermittent dextrose boluses are necessary to prevent hypoglycemia; maintain euglycemia goal of 100 to 250 mg/dL (Ref). When discontinuing the insulin infusion, the hypoglycemic effects may last for at least 24 hours after it is stopped; continue to monitor glucose levels closely during that time.

Correct hypokalemia prior to initiation of insulin therapy; monitor serum potassium closely during insulin infusion and supplement as needed to maintain normal serum potassium levels (Ref). To avoid fluid overload, concentrate IV fluids, including dextrose boluses or supplementation as appropriate and as venous access allows (Ref).

Infants, Children, and Adolescents: IV: Initial loading dose: 0.5 to 1 unit/kg bolus followed by a continuous IV infusion starting at 0.5 to 1 unit/kg/hour; titrate to clinical response; it has been suggested if patient remains hypotensive after initial 30 minutes of infusion to increase rate to 2 units/kg/hour, with additional rate increases as needed to maintain normotension. In severe cases, higher doses (continuous infusion >10 units/kg/hour) may be necessary (Ref).

Hyperkalemia, treatment

Hyperkalemia, treatment: Limited data available: **Note:** Experts suggest using a ratio of 1 unit of insulin for every 5 g of dextrose (Ref).

Infants, Children, and Adolescents: IV: 0.1 unit/kg (maximum dose: 10 units/dose) combined with dextrose administered over 30 minutes (Ref). An alternate approach is dextrose bolus followed by 0.2 units of insulin per g of dextrose administered over 15 to 30 minutes then infused continuously as a similar amount per hour (Ref). In adults, the usual dose is 10 units of insulin mixed with 25 g of dextrose (50 mL of D50W) administered over 15 to 30 minutes (Ref).

Hyperosmolar hyperglycemic state

Hyperosmolar hyperglycemic state (HHS): Limited data available: Infants, Children, and Adolescents:

Note: Only regular IV insulin should be used. Insulin administration should be initiated when serum glucose concentration is no longer declining at a rate ≥ 50 mg/dL/hour with fluid administration alone; earlier initiation may be required in patients with severe ketosis and acidosis. Refer to institution-specific protocols where appropriate.

Continuous IV infusion: Initial: 0.025 to 0.05 units/kg/hour; titrate dose to achieve a decrease in serum glucose concentration at a rate of 50 to 75 mg/dL/hour; discontinue insulin if serum glucose concentration decreases >100 mg/dL/hour. Insulin boluses are not recommended (Ref).

Dosage adjustment for concomitant therapy: Significant drug interactions exist, requiring dose/frequency adjustment or avoidance. Consult drug interactions database for more information.

Dosing: Kidney Impairment: Pediatric

There are no dosage adjustments provided in manufacturer's labeling; insulin requirements are reduced due to changes in insulin clearance or metabolism; monitor blood glucose closely. Based on experience in adult patients, dosage adjustment may be necessary.

Dosing: Liver Impairment: Pediatric

There are no dosage adjustments provided in manufacturer's labeling; insulin requirements may be reduced due to changes in insulin clearance or metabolism; monitor blood glucose closely.

Adverse Reactions

The following adverse drug reactions and incidences are derived from product labeling unless otherwise specified.

Frequency not defined:

Cardiovascular: Peripheral edema

Dermatologic: Injection site pruritus

Endocrine & metabolic: Amyloidosis (localized at injection site), hypoglycemia, hypokalemia, weight gain

Hypersensitivity: Anaphylaxis, hypersensitivity reaction

Immunologic: Immunogenicity

Local: Erythema at injection site, hypertrophy at injection site, lipoatrophy at injection site, swelling at injection site

Contraindications

Hypersensitivity to regular insulin or any component of the formulation; during episodes of hypoglycemia.

Warnings/Precautions

Concerns related to adverse effects:

- Glycemic control: Hyper- or hypoglycemia may result from changes in insulin strength, manufacturer, type, and/or administration method. The most common adverse effect of insulin is hypoglycemia. The timing of hypoglycemia differs among various insulin formulations. Hypoglycemia may result from changes in meal pattern (eg, macronutrient content, timing of meals), changes in the level of physical activity, increased work or exercise without eating, or changes to coadministered medications. Use of long-acting insulin preparations (eg, insulin degludec, insulin detemir, insulin glargine) may delay recovery from hypoglycemia. Patients with renal or hepatic impairment may be at a higher risk. Symptoms differ in patients and may change over time in the same patient; awareness may be less pronounced in those with long-standing diabetes, diabetic nerve disease, patients taking beta-blockers, or in those who experience recurrent hypoglycemia. Profound and

prolonged episodes of hypoglycemia may result in convulsions, unconsciousness, temporary or permanent brain damage, or even death. Insulin requirements may be altered during illness, emotional disturbances, or other stressors. Instruct patients to use caution with ethanol; may increase risk of hypoglycemia.

- **Hypersensitivity:** Severe, life-threatening, generalized allergic reactions, including anaphylaxis, may occur. If hypersensitivity reactions occur, discontinue therapy, treat the patient with supportive care and monitor until signs and symptoms resolve.
- **Hypokalemia:** Insulin (especially IV insulin) causes a shift of potassium from the extracellular space to the intracellular space, possibly producing hypokalemia. If left untreated, hypokalemia may result in respiratory paralysis, ventricular arrhythmia, and even death. Use with caution in patients at risk for hypokalemia (eg, loop diuretic use). Monitor serum potassium frequently with IV insulin use and supplement potassium when necessary.

Disease-related concerns:

- **Bariatric surgery:**
 - Presurgical assessment of the indication for use, symptoms, and goals of therapy should be documented to enable postsurgical assessment. Insulin therapy is the preferred treatment for diabetes in the early postoperative period; however, due to wide interpatient variability close monitoring and dosage adjustments are required to maintain target blood glucose levels and avoid hypoglycemia (AACE/ASMBS/OMA/ASA [Mechanick 2020]). Studies found that prior to discharge persons with type 1 diabetes required an average insulin dose reduction of 31% to 50% and patients with type 2 diabetes required a 61% to 87% dose reduction (Crujisen 2014; Diemer 2017; Kassel 2022; Middelbeek 2014; Vilarrasa 2017; Wirunsawanya 2021). Furthermore, 1 year following surgery 60% of persons with type 2 diabetes were in remission and able to discontinue insulin therapy (Ardestani 2015; Arterburn 2013; Buchwald 2009; Schauer 2003; Varban 2022).
 - DPP4 inhibitors, glucagon-like peptide-1 agonists, insulin, and metformin are preferred for outpatient glucose management following bariatric surgery (AACE/ASMBS/OMA/ASA [Mechanick 2020]). Consider alternatives to insulin for glucose management in type 2 diabetes due to the risk of hypoglycemia and weight gain associated with long-term insulin therapy (AACE/ASMBS/OMA/ASA [Mechanick 2020]; Apovian 2015). Monitor and adjust the insulin dose for hypoglycemia, hyperglycemia, or weight gain in persons requiring insulin, including persons with type 1 diabetes.
- **Cardiac disease:** Concurrent use with peroxisome proliferator-activated receptor (PPAR)-gamma agonists, including thiazolidinediones, may cause dose-related fluid retention and lead to or exacerbate heart failure, particularly when used in combination with insulin. If PPAR-gamma agonists are prescribed, monitor for signs and symptoms of heart failure. If heart failure develops, consider PPAR-gamma agonist dosage reduction or therapy discontinuation.
- **Hepatic impairment:** Use with caution in patients with hepatic impairment; increased risk of hypoglycemia. Dosage requirements may be reduced and patients may require more frequent dose adjustment and glucose monitoring.
- **Renal impairment:** Use with caution in patients with renal impairment; increased risk of hypoglycemia. Dosage requirements may be reduced and patients may require more frequent dose adjustment and glucose monitoring.

Special populations:

- **Hospitalized patients:** Prolonged use of a correctional-only (ie, sliding scale) insulin regimen in the inpatient setting is strongly discouraged. In the critical care setting, continuous IV insulin infusion has been shown to best achieve glycemic targets. In noncritically ill patients with either poor oral intake or taking nothing by mouth, basal insulin use is preferred, with correctional doses (insulin regular or rapid-acting insulin) as needed. In noncritically ill patients with adequate nutritional intake, a combination of basal insulin along with nutritional and correctional components (insulin regular or rapid-acting insulin) is preferred. An effective insulin regimen will achieve the goal glucose range without the risk of severe hypoglycemia. A blood glucose value <70 mg/dL should prompt a treatment regimen review and change, if necessary, to prevent further hypoglycemia (ADA 2025).

Dosage form specific issues:

- Multiple-dose injection pens: According to the Centers for Disease Control and Prevention (CDC), pen-shaped injection devices should never be used for more than one person (even when the needle is changed) because of the risk of infection. The injection device should be clearly labeled with individual patient information to ensure that the correct pen is used (CDC 2012).
- Product variation: Human insulin differs from animal-source insulin. Any change of insulin should be made cautiously; changing manufacturers, type, and/or method of manufacture may result in the need for a change of dosage. Verify product label prior to administration to prevent medication errors.
- U-500 regular insulin: **U-500 regular insulin is a concentrated insulin formulation which contains 500 units of insulin per mL and is intended for SUBQ administration only; do not administer IV or IM.** U-500 regular insulin is generally not recommended for use in an insulin pump but may be used in select patients under the supervision of an experienced provider (AACE [Grunberger 2010]). Prescribe only to patients who require >200 units of insulin per day. Doses from a U-500 regular insulin vial should be drawn up only with a dedicated U-500 insulin syringe. Do not mix or dilute U-500 regular insulin with other insulin formulations. Insulin U-500 also has a delayed onset and longer duration of action compared to regular insulin U-100, and has both prandial and basal properties (ADA 2025). Do not perform dose conversions when using the KwikPen, the dose window shows the number of units to be injected. Do not transfer insulin from the KwikPen to a syringe for administration.

Other warnings/precautions:

- Patient education: Diabetes self-management education is essential to maximize the effectiveness of therapy.

Dosage Forms: US

Excipient information presented when available (limited, particularly for generics); consult specific product labeling. [DSC] = Discontinued product

Solution, Injection:

HumuLIN R: 100 units/mL (3 mL, 10 mL) [contains metacresol]

NovoLIN R: 100 units/mL (10 mL) [contains metacresol]

NovoLIN R ReliOn: 100 units/mL (10 mL) [contains metacresol]

Solution, Intravenous [preservative free]:

Myxredlin: 100 units/100 mL in NaCl 0.9% (100 mL)

Solution, Subcutaneous:

HumuLIN R U-500 (CONCENTRATED): 500 units/mL (20 mL [DSC]) [contains metacresol]

Solution Pen-injector, Injection:

NovoLIN R FlexPen: 100 units/mL (3 mL) [contains metacresol]

NovoLIN R FlexPen ReliOn: 100 units/mL (3 mL) [contains metacresol]

Solution Pen-injector, Subcutaneous:

HumuLIN R U-500 KwikPen: 500 units/mL (3 mL) [contains metacresol]

Generic Equivalent Available: US

No

Pricing: US

Solution (HumuLIN R Injection)

100 units/mL (per mL): \$5.35

Solution (Myxredlin Intravenous)

100UT/100ML 0.9% (per mL): \$0.42

Solution (NovoLIN R Injection)

100 units/mL (per mL): \$5.78

Solution (NovoLIN R ReliOn Injection)

100 units/mL (per mL): \$5.78

Solution Pen-injector (HumuLIN R U-500 KwikPen Subcutaneous)

500 units/mL (per mL): \$114.84

Solution Pen-injector (NovoLIN R FlexPen Injection)

100 units/mL (per mL): \$7.29

Solution Pen-injector (NovoLIN R FlexPen ReliOn Injection)

100 units/mL (per mL): \$7.29

Disclaimer: A representative AWP (Average Wholesale Price) price or price range is provided as reference price only. A range is provided when more than one manufacturer's AWP price is available and uses the low and high price reported by the manufacturers to determine the range. The pricing data should be used for benchmarking purposes only, and as such should not be used alone to set or adjudicate any prices for reimbursement or purchasing functions or considered to be an exact price for a single product and/or manufacturer. Medi-Span expressly disclaims all warranties of any kind or nature, whether express or implied, and assumes no liability with respect to accuracy of price or price range data published in its solutions. In no event shall Medi-Span be liable for special, indirect, incidental, or consequential damages arising from use of price or price range data. Pricing data is updated monthly.

Dosage Forms: Canada

Excipient information presented when available (limited, particularly for generics); consult specific product labeling.

Solution, Injection:

HumuLIN R: 100 units/mL (3 mL, 10 mL)

NovoLIN GE Toronto: 100 units/mL (10 mL) [contains metacresol]

Solution, Intravenous:

Myxrealin: 100 units/100 mL in NaCl 0.9% (100 mL)

Solution Cartridge, Injection:

NovoLIN GE Toronto Penfill: 100 units/mL (1.5 mL, 3 mL)

Solution Pen-injector, Injection:

HumuLIN R KwikPen: 100 units/mL (3 mL) [contains metacresol]

Solution Pen-injector, Subcutaneous:

Entuzity KwikPen: 500 units/mL (3 mL) [contains metacresol]

Administration: Adult

SUBQ administration: Do not use if solution is viscous or cloudy; use only if clear and colorless. Regular insulin should be administered ~30 minutes before a meal. Cold injections should be avoided. SUBQ administration is usually made into the thighs, arms, buttocks, or abdomen; rotate injection sites within the same region to avoid lipodystrophy or localized cutaneous amyloidosis. Rotating from an injection site where lipodystrophy/cutaneous amyloidosis is present to an unaffected site may increase risk of hypoglycemia.

U-100 regular insulin: When mixing U-100 regular insulin with other preparations of insulin, regular insulin should be drawn into syringe first.

FlexPen: Prime the needle before each injection with 2 units of insulin. Once injected, hold the needle in the skin for at least 6 seconds after the dose dial has returned to 0 units before removing the needle to ensure the full dose has been administered.

U-500 regular insulin (concentrated): Do not dilute or mix U-500 regular insulin.

Vials: U-500 regular insulin vials are to be used only in conjunction with a dedicated U-500 insulin syringe; dosage conversion is not required with the U-500 syringe. Only in cases where the U-500 insulin syringe is not available, a U-100 insulin syringe or a tuberculin syringe may be necessary. When using a U-100 syringe or a tuberculin syringe to deliver Humulin R U-500 (from vial), a conversion step is required to ensure the correct amount of Humulin R U-500 is drawn up in the syringe. To avoid dosing errors when using a U-100 insulin syringe, the prescribed dose should be written in actual insulin units and as unit markings on the U-100 insulin syringe (eg, Humulin R U-500 50 units = 10 units on a U-100 insulin syringe). To avoid dosing errors when using a tuberculin syringe, the prescribed dose should be written in actual insulin units and as a volume (eg, Humulin R U-500 50 units = 0.1 mL on a tuberculin syringe).

Humulin KwikPen (U-500): Do not perform dose conversions when using the KwikPen; the dose window shows the number of units to be injected. The KwikPen dials by 5-unit increments, resulting in delivery of doses in 5-unit increments. Do not transfer KwikPen insulin into a syringe for administration. Prime the needle before each injection with 5 units of insulin. Once the dose is injected, hold the needle in the skin for ~5 seconds after the dose dial has returned to 0 units before removing the needle to ensure the full dose has been administered.

Entuzity KwikPen (U-500) [Canadian product]: Do not perform dose conversions when using the KwikPen; the dose window shows the number of units to be injected. The KwikPen dials by 5-unit increments, resulting in delivery of doses in 5-unit increments. Do not transfer KwikPen insulin into a syringe for administration.

Continuous SUBQ insulin infusion: Novolin regular insulin (U-100) is not recommended for use in external SUBQ insulin infusion pump due to precipitation concerns (manufacturer labeling). Concentrated U-500 regular insulin is generally not recommended for use in an insulin pump but may be used in select patients under the supervision of an experienced provider (Ref).

IV administration: Do not administer concentrated U-500 regular insulin or mixtures of insulin formulations IV. Do not use if solution is viscous or cloudy; use only if clear and colorless. U-100 regular insulin may be administered IV with close monitoring of blood glucose and serum potassium; appropriate medical supervision is required. **Note:** A 100 units/100 mL premixed solution is also available for IV infusion.

IV infusions: To minimize insulin adsorption to plastic IV tubing: Insulin loss will occur by adsorption to plastic (ie, PVC, polyethylene, polyolefin, polypropylene) IV containers and tubing (Ref). Therefore, flush the IV tubing with a priming infusion of 20 mL from the insulin infusion, whenever a new IV tubing set is added to the insulin infusion container (Ref).

If insulin is required prior to the availability of the insulin infusion, regular insulin should be administered by IV push injection.

Because of insulin adsorption to plastic IV tubing or infusion bags, the actual amount of insulin being administered via IV infusion could be substantially less than the apparent amount. Therefore, adjustment of the IV infusion rate should be based on effect and not solely on the apparent insulin dose. The apparent dose may be used as a starting point for determining the subsequent SUBQ dosing regimen (Ref); however, the transition to SUBQ administration requires continuous medical supervision, frequent monitoring of blood glucose, and careful adjustment of therapy. In addition, SUBQ basal insulin should be administered ≥ 2 to 4 hours before discontinuing IV insulin infusion to prevent hyperglycemia (Ref).

Preparation for Administration: Adult

For SUBQ administration: *Humulin R U-100:* May be diluted with manufacturer's sterile diluent down to a concentration of 50 units/mL (U-50) or 10 units/mL (U-10).

For IV infusion: Note: A 100 units/100 mL premixed solution is also available for IV infusion.

Humulin R U-100: Dilute in NS to concentrations of 0.1 to 1 unit/mL. **Do not administer concentrated U-500 regular insulin IV.**

Novolin R: Dilute in NS, D5W, or D10W with 40 mEq/L potassium chloride to concentrations of 0.05 to 1 unit/mL.

Calcium channel blocker or beta-blocker toxicity (off-label use):

Note: Final diluted insulin solution concentrations greater than 1 unit/mL may be required to prevent fluid overload in patients receiving high doses. A standard, distinct insulin concentration designated for use in calcium channel blocker or beta-blocker overdose may help to avoid medication errors (Ref).

Add 800 units (8 mL of a 100 units/mL solution) of regular insulin to 42 mL NS to a final concentration of 16 units/mL (Ref).

Usual Infusion Concentrations: Adult

IV infusion: 100 units in 100 mL (concentration: 1 unit/mL) of NS.

IV infusion for calcium channel blocker or beta-blocker toxicity: 800 units in 42 mL of NS (concentration: 16 units/mL) (Ref).

Administration: Pediatric

Parenteral: Do not use if solution is viscous or cloudy; use only if clear and colorless.

SUBQ: Administer ~30 minutes before meals. Cold injections should be avoided. Administration is usually made into the subcutaneous fat of the thighs, arms, buttocks, or abdomen; rotate injection sites within the same region to avoid lipodystrophy or localized cutaneous amyloidosis. Rotating from an injection site where lipodystrophy/cutaneous amyloidosis is present to an unaffected site may increase risk of hypoglycemia. Per the manufacturer, Novolin regular insulin is not recommended for use in external SUBQ insulin infusion pump due to precipitation concerns.

U-100 regular insulin:

Vial: When mixing U-100 regular insulin with other insulin preparations, regular insulin should be drawn into the syringe first. While not preferred, regular insulin may be infused SUBQ by external insulin pump (eg, when rapid-acting insulin not available) (Ref); however, when used in an external pump, it is not recommended to be diluted with other solutions.

Prefilled pens: Novolin R FlexPen and Novolin R FlexPen ReliOn are available in a concentration of 100 units/mL. Devices are designed to display the actual insulin units administered (no dosage conversion needed) and will administer up to 60 units per injection in 1-unit increments. Prime the needle before each injection with 2 units of insulin (use a new needle for each injection); see manufacturer's labeling for specific procedure. Once primed, set dial to the appropriate dose, insert needle into clean skin, and activate device by holding the button down; continue to hold the button until the dose dial has returned to 0 units. After the insulin is injected, hold the needle in the skin for ≥ 6 seconds after the dose dial has returned to 0 units before removing the needle to ensure the full dose has been administered. Do not rub the area. If dose is >60 units, >1 injection will be required; split dose and administer in multiple injections. Do not mix other insulin formulations with insulin regular contained in a prefilled pen.

U-500 regular insulin (concentrated): Concentrated U-500 regular insulin is indicated only in patients requiring >200 units/day of insulin. Do not dilute or mix U-500 regular insulin.

Vial: **U -500 regular insulin vials are to be used only in conjunction with a dedicated U-500 insulin syringe; dosage conversion is not required with the U-500 syringe.** Only in cases where the U-500 insulin syringe is not available, a U-100 insulin syringe or a tuberculin syringe may be necessary. When using a U-100 syringe or a tuberculin syringe to deliver Humulin R U-500 (from

vial), a conversion step is required to ensure the correct amount of Humulin R U-500 is drawn up in the syringe. To avoid dosing errors when using a U-100 insulin syringe, the prescribed dose should be written in actual insulin units and as unit markings on the U-100 insulin syringe (eg, Humulin R U-500 50 units = 10 units on a U-100 insulin syringe). To avoid dosing errors when using a tuberculin syringe, the prescribed dose should be written in actual insulin units and as a volume (eg, Humulin R U-500 50 units = 0.1 mL on a tuberculin syringe).

Prefilled pen: Do not perform dose conversions when using the KwikPen; the dose window shows the number of units to be injected. The Humulin R KwikPen will administer up to 300 units per injection in 5-unit increments. Prime the needle before each injection with 5 units of insulin (use a new needle for each injection); see manufacturer's labeling for specific procedure. Once primed, set dial to the appropriate dose, insert needle into clean skin, and activate device by holding the button down; continue to hold the button until the dose dial has returned to 0 units. After the insulin is injected, hold the needle in the skin for 5 seconds after the dose dial has returned to 0 units before removing the needle to ensure the full dose has been administered. Do not rub the area. If dose is >300 units, >1 injection will be required; split dose and administer in multiple injections. Do not transfer KwikPen insulin into a syringe for administration.

IM: *U-100 regular insulin:* May be administered IM in selected clinical situations; close monitoring of blood glucose and serum potassium as well as medical supervision is required. **Do not administer U-500 regular insulin IM.**

IV: *U-100 regular insulin:* Preferred insulin formulation approved for IV administration; requires close monitoring of blood glucose and serum potassium; appropriate medical supervision. Do not administer U-500 regular insulin or mixtures of insulin formulations intravenously.

Continuous IV Infusion: *U-100 regular insulin:* To minimize insulin adsorption to plastic IV tubing: Insulin loss will occur by adsorption to plastic (ie, PVC, polyethylene, polyolefin, polypropylene) IV containers and tubing (Ref). Therefore, flush IV tubing with a priming volume of 20 mL from the insulin infusion, whenever a new IV tubing set is added to the insulin infusion container (Ref). Studies examining this issue in neonates suggest that flushing the IV tubing prior to administration reduces adsorption and provides improved and more predictable insulin delivery; however, the combination of flushing along with preconditioning (waiting a predefined time after flushing the IV line before infusing) provides the greatest reduction in insulin adsorption; wait times for preconditioning varied among studies from 20 to 60 minutes; flush volumes varied and were as high as 20 mL (Ref).

High-dose insulin therapy (for treating calcium channel blocker or beta-blocker toxicity): *U-100 regular insulin:* Administer as a continuous IV infusion. Additional precautions should be implemented to ensure accurate infusion pump settings, with particular attention to infusion concentration required (Ref). Refer to institution-specific protocols where appropriate.

Because of adsorption to plastic IV tubing or infusion bags, the actual amount of insulin being administered could be substantially less than the apparent amount. Therefore, adjustment of the IV infusion rate should be based on the effect and not solely on the apparent insulin dose. The apparent dose may be used as a starting point for determining the subsequent SUBQ dosing regimen (Ref); however, the transition to SUBQ administration requires continuous medical supervision, frequent monitoring of blood glucose, and careful adjustment of therapy.

Preparation for Administration: Pediatric

Parenteral:

SubQ: Regular insulin may be mixed with other insulins; when mixing regular insulin with other insulin preparations, regular insulin should be drawn into syringe first.

Humulin R U-100: May be diluted with the manufacturer's sterile diluent down to a concentration of 10 units/mL (U-10) (Ref).

IV infusion: Only use U-100 preparations to prepare infusions. **Note:** A premix solution of 100 units/100 mL NS (1 unit/mL) is also available for IV infusion which may be appropriate in some pediatric patients.

Diabetic ketoacidosis (DKA), hyperosmolar hyperglycemic state (HHS):

Humulin R U-100: May be diluted in NS or D5W to concentrations of 0.1 to 1 unit/mL.

Novolin R: May be diluted in NS, D5W, **or** D10W with 40 mEq/L potassium chloride at concentrations of 0.05 to 1 unit/mL.

Note: ISMP and Vermont Oxford Network recommend a standard concentration of 0.1 or 0.5 units/mL for neonates (Ref); for pediatric patients a standard concentration of 0.2 or 1 units/mL is recommended (Ref).

Calcium channel blocker or beta-blocker overdose/toxicity (high-dose insulin therapy):

Note: To prevent medication errors, a standard and unique insulin infusion concentration and preparation (eg, unique final volume) specific for use in calcium channel blocker or beta-blocker toxicity is recommended. Due to the higher rates of infusion required and to prevent fluid overload, a more concentrated insulin solution (eg, 16 units/mL or 20 units/mL) may be necessary (Ref).

Concentrated insulin infusion: U-100 preparation: To prepare 16 units/mL, add 800 units (8 mL of 100 units/mL) regular insulin to 42 mL NS; final concentration of 16 units/mL has been suggested (Ref).

Usual Infusion Concentrations: Pediatric

IV infusion: 0.1 unit/mL, 0.2 unit/mL, 0.5 unit/mL, or 1 unit/mL.

Storage/Stability

Humulin R U-100: Store **unopened** vials under refrigeration between 2°C and 8°C (36°F and 46°F) until expiration date or at room temperature ≤30°C (≤86°F) for 31 days; do not freeze and do not use if the vial has been frozen; keep away from heat and sunlight. Store **in-use** vials under refrigeration between 2°C and 8°C (36°F and 46°F) or at room temperature ≤30°C (≤86°F) and use within 31 days.

Humulin R U-500:

Vials: Store **unopened** vials under refrigeration between 2°C and 8°C (36°F and 46°F) until expiration date or at room temperature ≤30°C (≤86°F) for 40 days; do not shake; protect from heat and light; do not freeze and do not use if the vial has been frozen. Store **in-use** vials under refrigeration between 2°C and 8°C (36°F and 46°F) or at room temperature ≤30°C (≤86°F) and use within 40 days.

KwikPen: Store **unopened** prefilled pens under refrigeration between 2°C and 8°C (36°F and 46°F) until expiration date or at room temperature ≤30°C (≤86°F) for 28 days; protect from heat and light; do not freeze and do not use if the pen has been frozen. Store **in-use** prefilled pens at room temperature ≤30°C (≤86°F) and use within 28 days; do not refrigerate.

Myxrealin: Store at 2°C to 8°C (36°F to 46°F) in original carton to protect from light. Do not freeze or shake. May be stored at room temperature (<25°C [$<77^{\circ}\text{F}$]) for ≤30 days; do not return to refrigerator once stored at room temperature; discard after 30 days.

Novolin R:

Vials: Store **unopened** vials under refrigeration between 2°C and 8°C (36°F and 46°F) until expiration date or at room temperature ≤25°C (≤77°F) for 42 days; do not freeze; keep away from heat and sunlight. Store **in-use** vials at room temperature ≤25°C (≤77°F) and use within 42 days; do not refrigerate.

FlexPen: Store **unopened** prefilled pens under refrigeration between 2°C and 8°C (36°F and 46°F) until expiration date or at room temperature ≤30°C (≤86°F) for 28 days; do not use if frozen; keep away from heat and light. Store **in-use** prefilled pens at room temperature ≤30°C (≤86°F) and use within 28 days.

Canadian labeling (not in US labeling): All products: Store **unopened** vials, cartridges, and pre-filled pens under refrigeration between 2°C and 8°C (36°F and 46°F) until expiration date; do not freeze; keep away from heat and sunlight. Store **in-use** Humulin vials, cartridges, and prefilled pens at room temperature <25°C (<77°F) and use within 4 weeks. Store **in-use** Novolin GE vials,

cartridges, and prefilled pens at room temperature <25°C (<77°F) for vials or <30°C (<86°F) for prefilled pens/cartridges and use within 1 month; do not refrigerate. Store **in-use** Entuzity KwikPen prefilled pens at room temperature ≤30°C (≤86°F) and use within 4 weeks; do not refrigerate.

Diluted solution for SUBQ administration: *Humulin R U-100*: Diluted insulin solution may be stored at 30°C (86°F) for ≤14 days or at 5°C (41°F) for ≤28 days.

According to the manufacturer, do not store diluted insulin in a plastic syringe; store unused sterile diluent at room temperature; once in use, the sterile diluent vial should be used within 28 days (data on file [Eli Lilly 2018]).

For IV infusion: Note: A 100 units/100 mL premixed solution is also available for IV infusion.

Humulin R U-100: Stable for 48 hours at room temperature or for 48 hours under refrigeration followed by 48 hours at room temperature.

Novolin R: Stable for 24 hours at room temperature

Note: After dilution of 100 units of regular human insulin (product not specified) in 100 mL of 0.9% NaCl (PVC bag), the solution is stable under refrigeration between 2°C and 8°C (36°F to 46°F) for ≤336 hours (Rocchio 2013).

After dilution of 800 units of regular insulin (8 mL of a 100 units/mL solution) in 42 mL of 0.9% NaCl (PVC bag) to a final concentration of 16 units/mL, the solution is stable under refrigeration at 4°C (39°F) or at room temperature at 25°C (77°F) for 14 days (Laskey 2016).

Use: Labeled Indications

Diabetes mellitus, types 1 and 2, treatment: Treatment of type 1 diabetes mellitus and type 2 diabetes mellitus to improve glycemic control.

Note: Concentrated U-500 regular insulin is indicated only in patients requiring more than 200 units of insulin per day.

Use: Off-Label: Adult

Cadaveric organ recovery (hormonal resuscitation); Calcium channel blocker or beta-blocker toxicity; Diabetic ketoacidosis; Diabetes mellitus, gestational; Hyperglycemia, hospitalized patients; Hyperkalemia, severe/emergent; Hyperosmolar hyperglycemic state

Medication Safety Issues

Sound-alike/look-alike issues:

HumuLIN R may be confused with HumaLOG, Humira, HumuLIN 70/30, HumuLIN N, NovoLIN 70/30, NovoLIN R, NovoLOG

HumuLIN R U-500 may be confused with Humulin R U-100

NovoLIN R may be confused with HumuLIN R, NovoLIN 70/30, NovoLIN N, NovoLOG

Insulin may be confused with influenza virus vaccine. Medication errors have occurred when insulin was inadvertently administered instead of influenza virus vaccine. These products are refrigerated and may be stored in close proximity to each other.

High alert medication:

The Institute for Safe Medication Practices (ISMP) includes this medication among its list of drug classes (insulins, all formulations and strengths, SUBQ, IV; with special emphasis on U-500 insulin) which have a heightened risk of causing significant patient harm when used in error (ISMP List of High-Alert Medications in Acute Care Settings ©ISMP 2024; ISMP List of High-Alert Medications in Community/Ambulatory Care Settings ©ISMP 2021; High-Alert Medications in Long-Term Care (LTC) Settings ©ISMP 2021). Due to the number of insulin preparations, it is essential to identify/clarify the type of insulin to be used.

Older Adult: High-Risk Medication:

Beers Criteria: Insulin (short- or rapid-acting insulin products used for sliding scale) is identified in the Beers Criteria as a potentially inappropriate medication to be avoided in patients 65 years and older (independent of diagnosis or condition) due to higher risk of hypoglycemia associated with sliding scale insulin without improvements in hyperglycemia, regardless of care setting. Avoid insulin regimens that only use short- or rapid-acting insulins dosed based on current blood glucose levels in the absence of basal or long-acting insulin; recommendation does not apply to regimens containing basal or long-acting insulin (Beers Criteria [AGS 2023]).

Administration issues:

Concentrated solutions (eg, U-500) should not be available in patient care areas. U-500 regular insulin should be stored, dispensed, and administered separately from U-100 regular insulin. U-500 insulin vials are to be used in conjunction only with a dedicated U-500 insulin syringe. Though no longer recommended by the manufacturer, in cases where the U-500 insulin syringe is not available, a U-100 insulin syringe or a tuberculin syringe may be used (**conversion step required to ensure correct dose is drawn up**). For patients who receive U-500 insulin in the hospital setting, highlighting the strength prominently on the patient's medical chart and medication record may help to reduce dispensing errors. For patients being discharged from the hospital and/or in the outpatient setting, a U-500 insulin syringe is available by prescription only and should be prescribed/dispensed in conjunction with U-500 insulin vials.

Other safety concerns:

Cross-contamination may occur if insulin pens are shared among multiple patients. Steps should be taken to prohibit sharing of insulin pens.

Metabolism/Transport Effects

None known.

Drug Interactions

(For additional information: Launch drug interactions program)

Note: Interacting drugs may **not be individually listed below** if they are part of a group interaction (eg, individual drugs within "CYP3A4 Inducers [Strong]" are NOT listed). For a complete list of drug interactions by individual drug name and detailed management recommendations, use the drug interactions program by clicking on the "Launch drug interactions program" link above.

Alcohol (Ethyl): May increase or decrease hypoglycemic effects of Insulin. *Risk C: Monitor*

Alpha-Glucosidase Inhibitors: May increase hypoglycemic effects of Insulin. Management: Consider a decrease in insulin dose when initiating therapy with an alpha-glucosidase inhibitor and monitor patients for hypoglycemia. *Risk D: Consider Therapy Modification*

Alpha-Lipoic Acid: May increase hypoglycemic effects of Antidiabetic Agents. *Risk C: Monitor*

Androgens: May increase hypoglycemic effects of Agents with Blood Glucose Lowering Effects. *Risk C: Monitor*

Antidiabetic Agents: May increase hypoglycemic effects of Hypoglycemia-Associated Agents. *Risk C: Monitor*

Beta-Blockers (Beta1 Selective): May increase adverse/toxic effects of Antidiabetic Agents. Specifically, beta-blockers may mask the hypoglycemic symptoms of antidiabetic agents. *Risk C: Monitor*

Beta-Blockers (Nonselective): May increase hypoglycemic effects of Insulin. Beta-Blockers (Nonselective) may decrease therapeutic effects of Insulin. *Risk C: Monitor*

Bortezomib: May increase therapeutic effects of Antidiabetic Agents. Bortezomib may decrease therapeutic effects of Antidiabetic Agents. *Risk C: Monitor*

Chlorprothixene: May increase hypoglycemic effects of Insulin. *Risk C: Monitor*

Clarithromycin: May increase hypoglycemic effects of Insulin. *Risk C: Monitor*

CloNIDine: May increase or decrease hypoglycemic effects of Insulin. *Risk C: Monitor*

Dipeptidyl Peptidase-IV Inhibitors: May increase hypoglycemic effects of Insulin. Management: Consider a decrease in insulin dose when initiating therapy with a dipeptidyl peptidase-IV inhibitor and monitor patients for hypoglycemia. *Risk D: Consider Therapy Modification*

Direct Acting Antiviral Agents (HCV): May increase hypoglycemic effects of Antidiabetic Agents. *Risk C: Monitor*

Edetate CALCIUM Disodium: May increase hypoglycemic effects of Insulin. *Risk C: Monitor*

Etilefrine: May decrease therapeutic effects of Antidiabetic Agents. *Risk C: Monitor*

Glucagon-Like Peptide-1 Agonists: May increase hypoglycemic effects of Insulin. Management: Consider insulin dose reductions when used in combination with glucagon-like peptide-1 agonists. Monitor patients for hypoglycemia. *Risk D: Consider Therapy Modification*

Guanethidine: May increase hypoglycemic effects of Antidiabetic Agents. *Risk C: Monitor*

Herbal Products with Glucose Lowering Effects: May increase hypoglycemic effects of Hypoglycemia-Associated Agents. *Risk C: Monitor*

Hyperglycemia-Associated Agents: May decrease therapeutic effects of Antidiabetic Agents. *Risk C: Monitor*

Hypoglycemia-Associated Agents: Antidiabetic Agents may increase hypoglycemic effects of Hypoglycemia-Associated Agents. *Risk C: Monitor*

Hypoglycemia-Associated Agents: May increase hypoglycemic effects of Hypoglycemia-Associated Agents. *Risk C: Monitor*

Liraglutide: May increase hypoglycemic effects of Insulin. Management: Consider reducing the liraglutide dose if coadministered with insulin. Prescribing information for the Saxenda brand of liraglutide recommends a dose decrease of 50%. Monitor blood glucose for hypoglycemia. *Risk D: Consider Therapy Modification*

Lithium: May increase or decrease hypoglycemic effects of Insulin. *Risk C: Monitor*

Macimorelin: Coadministration of Insulin and Macimorelin may alter diagnostic results. *Risk X: Avoid*

Maitake: May increase hypoglycemic effects of Agents with Blood Glucose Lowering Effects. *Risk C: Monitor*

Metreleptin: May increase hypoglycemic effects of Insulin. Management: Insulin dosage adjustments (including potentially large decreases) may be required to minimize the risk for hypoglycemia with concurrent use of metreleptin. Monitor closely for signs and symptoms of hypoglycemia. *Risk D: Consider Therapy Modification*

Monoamine Oxidase Inhibitors: May increase hypoglycemic effects of Agents with Blood Glucose Lowering Effects. *Risk C: Monitor*

Pegvisomant: May increase hypoglycemic effects of Agents with Blood Glucose Lowering Effects. *Risk C: Monitor*

Pioglitazone: May increase adverse/toxic effects of Insulin. Specifically, the risk for hypoglycemia, fluid retention, and heart failure may be increased with this combination. Management: If insulin is combined with pioglitazone, consider insulin dose reductions to avoid hypoglycemia. Monitor patients for fluid retention and signs/symptoms of heart failure, and consider pioglitazone dose reduction or discontinuation if heart failure occurs *Risk D: Consider Therapy Modification*

Pramlintide: May increase hypoglycemic effects of Insulin. Management: Upon initiation of pramlintide, decrease mealtime insulin dose by 50% to reduce the risk of hypoglycemia. Monitor blood glucose frequently and individualize further insulin dose adjustments based on glycemic control. *Risk D: Consider Therapy Modification*

Prothionamide: May increase hypoglycemic effects of Agents with Blood Glucose Lowering Effects. *Risk C: Monitor*

Quinolones: May increase hypoglycemic effects of Agents with Blood Glucose Lowering Effects. Quinolones may decrease therapeutic effects of Agents with Blood Glucose Lowering Effects. Specifically, if an agent is being used to treat diabetes, loss of blood sugar control may occur with quinolone use. *Risk C: Monitor*

Reproterol: May decrease therapeutic effects of Antidiabetic Agents. *Risk C: Monitor*

Ritodrine: May decrease therapeutic effects of Antidiabetic Agents. *Risk C: Monitor*

Rosiglitazone: Insulin may increase adverse/toxic effects of Rosiglitazone. Specifically, the risk of fluid retention, heart failure, and hypoglycemia may be increased with this combination. Management: Consider alternatives when possible; labeling states that the use of rosiglitazone and insulin is not recommended. If combined, monitor for evidence of fluid retention, heart failure, and hypoglycemia. *Risk D: Consider Therapy Modification*

Salicylates: May increase hypoglycemic effects of Agents with Blood Glucose Lowering Effects. *Risk C: Monitor*

Selective Serotonin Reuptake Inhibitor: May increase hypoglycemic effects of Agents with Blood Glucose Lowering Effects. *Risk C: Monitor*

Sodium-Glucose Cotransporter 2 (SGLT2) Inhibitors: May increase hypoglycemic effects of Insulin. Management: Consider a decrease in insulin dose when initiating therapy with a sodium-glucose cotransporter 2 inhibitor and monitor patients for hypoglycemia. *Risk D: Consider Therapy Modification*

Thiazide and Thiazide-Like Diuretics: May decrease therapeutic effects of Antidiabetic Agents. *Risk C: Monitor*

Reproductive Considerations

Family planning is recommended as part of routine diabetic care for patients taking glucose lowering medications. Glucose control near physiologic levels is recommended prior to conception due to the risk of fetal malformations associated with poor glycemic control in early pregnancy. Recommend contraception for patients with diabetes who are not currently planning a pregnancy (AACE [Blonde 2022]; ACOG 201 2018; ADA 2025).

Pregnancy Considerations

Outcome data following maternal use of regular insulin during pregnancy are available (Brown 2017; O'Neill 2017); use does not show an association with an increased risk of adverse pregnancy outcomes.

Poor glycemic control during pregnancy can be associated with adverse pregnancy outcomes including increased risks for congenital malformations, spontaneous abortion, perinatal mortality, large for gestational age/macrosomia, neonatal hypoglycemia, neonatal hyperbilirubinemia, neonatal respiratory distress syndrome, preeclampsia, and preterm delivery. Long term effects include an increased risk of impaired glucose tolerance and obesity in children exposed to hyperglycemia in utero. Risks increase with degree of maternal hyperglycemia. Appropriate treatment helps reduce the risk of adverse outcomes; therefore, prior to conception and throughout pregnancy, keep maternal blood glucose and HbA_{1c} as close to target goals as possible without causing significant hypoglycemia (AACE [Blonde 2022]; ACOG 201 2018; ADA 2025).

Due to pregnancy-induced physiologic changes, insulin requirements tend to increase as pregnancy progresses, requiring frequent monitoring and dosage adjustments. Insulin resistance increases at ~16 weeks gestation with total daily insulin requirements increasing through 36 weeks gestation. Daily doses of insulin may double when compared to those required prior to pregnancy. Following delivery, insulin requirements decrease rapidly. Patients treated for gestational diabetes may no longer require medication postpartum (ACOG 201 2018; ADA 2025).

Insulin is the recommended treatment for all patients with type 1 diabetes during pregnancy and the preferred treatment for type 2 diabetes and gestational diabetes mellitus when pharmacologic

therapy is needed (AACE [Blonde 2022]; ACOG 190 2018; ACOG 201 2018; ADA 2025). Rapid-acting insulin analogs are preferred over regular insulin when treatment is needed during pregnancy due to improved outcomes and increased adherence (AACE [Blonde 2022]; ACOG 198 2018; ACOG 201 2018). Regular insulin is an alternative option for managing diabetes during pregnancy when rapid-acting insulin analogs are not available or cannot be taken (AACE [Blonde 2022]). Regular insulin may be used intravenously for glycemic control during labor and in the management of diabetic ketoacidosis in pregnant patients (ACOG 201 2018).

Breastfeeding Considerations

Both exogenous and endogenous insulin are present in breast milk (study not conducted with this preparation) (Whitmore 2012).

Adverse events have not been reported in breastfeeding infants following maternal use of regular insulin for injection. Insulin is not systemically absorbed via breast milk but may provide local benefits to the infant GI tract (Anderson 2018).

Appropriate glycemic control helps establish lactation in patients with insulin dependent diabetes (Anderson 2018). Breastfeeding provides metabolic benefits to mothers with type 1, type 2, and gestational diabetes mellitus as well as their infants; therefore, breastfeeding is encouraged (ACOG 201 2018; ADA 2025).

Breastfeeding also influences maternal glucose tolerance; close monitoring of patients treated with insulin is recommended as dose adjustments may be required (ADA 2025; Anderson 2018). According to the manufacturer, the decision to breastfeed during therapy should consider the risk of infant exposure, the benefits of breastfeeding to the infant, and the benefits of treatment to the mother. Adjust the daily insulin dose and daily amount of carbohydrate consumption if patients with pregestational diabetes experience overnight hypoglycemia following night-time breastfeeding (Ringholm 2024).

Dietary Considerations

Individualized medical nutrition therapy (MNT) based on ADA recommendations is an integral part of therapy.

Monitoring Parameters

Calcium channel blocker or beta-blocker toxicity (off-label use): Monitor blood glucose levels every 15 to 60 minutes, with frequent attention to blood glucose levels during the first hour of therapy. Serum electrolytes, including potassium, should be measured hourly until patient is stable, then monitor serum potassium levels every 1 to 2 hours and serum electrolyte measurement every 4 to 6 hours once stable. Consider serial ECG monitoring (Engebretsen 2011; Krenz 2018).

Diabetes mellitus and hyperglycemia: Blood glucose (individualize frequency based on treatment regimen, hypoglycemia risk, and other patient-specific factors) (ADA 2025); electrolytes; renal function; hepatic function; weight.

Gestational diabetes mellitus: Blood glucose 4 times daily (one fasting and three postprandial) until well controlled, then as appropriate (ACOG 190 2018).

Hospitalized patients: In patients who are eating, monitor blood glucose before meals and at bedtime; in patients who are not eating or are receiving continuous enteral feeds, monitor blood glucose every 4 to 6 hours (ADA 2025; ES [Umpierrez 2012]). More frequent monitoring may be required in some cases (eg, recurrent hypoglycemia, changes in nutrition, medication changes affecting glycemic control) (ES [Umpierrez 2012]).

Critically ill patients receiving an IV insulin infusion: Monitor blood glucose at least every 0.5 to 2 hours, with more frequent monitoring if glucose levels are unstable; serum potassium (ADA 2025; SCCM [Honarmand 2024]; manufacturer's labeling). Refer also to institutional protocols. **Note:** Arterial or venous whole blood sampling should be prioritized over point-of-care glucose testing (SCCM [Honarmand 2024]).

HbA_{1c}: Monitor at least twice yearly in patients who have stable glycemic control and are meeting treatment goals; monitor quarterly in patients in whom treatment goals have not been met, or with therapy change. **Note:** In patients prone to glycemic variability (eg, patients with insulin deficiency), or in patients whose HbA_{1c} is discordant with serum glucose levels or symptoms, consider evaluating HbA_{1c} in combination with blood glucose levels and/or a glucose management indicator (ADA 2025; KDIGO 2020).

Diabetic ketoacidosis (DKA) or hyperosmolar hyperglycemic state: Frequent monitoring (eg, every 1 to 4 hours) of serum electrolytes (eg, sodium, potassium, bicarbonate, phosphate), serum glucose, anion gap, venous pH and venous or capillary beta-hydroxybutyrate (for DKA), serum BUN, serum creatinine, serum osmolality, fluid status (eg, blood pressure, fluid intake/output, signs/symptoms of dehydration or fluid overload), and mental status. Refer also to institutional protocols (Hirsch 2025a; Hirsch 2025b; Umpierrez 2024).

Hyperkalemia: Serum potassium and glucose must be closely monitored to avoid hypokalemia, rebound hyperkalemia, and hypoglycemia.

Reference Range

Recommendations for glycemic control in adult patients with diabetes:

Nonpregnant patients (AACE [Samson 2023]; ADA 2025):

HbA_{1c}: <7% (a more aggressive [<6.5%] or less aggressive [<8%] HbA_{1c} goal may be targeted based on patient-specific characteristics).

Preprandial capillary blood glucose: 80 to 130 mg/dL (SI: 4.4 to 7.2 mmol/L); more or less stringent goals may be appropriate based on patient-specific characteristics.

Peak postprandial capillary blood glucose (~1 to 2 hours after a meal): <180 mg/dL (SI: <10 mmol/L); more or less stringent goals may be appropriate based on patient-specific characteristics.

Continuous glucose monitoring:

Time in range (70 to 180 mg/dL [SI: 3.9 to 10 mmol/L]): >70%.

Time above range (>180 mg/dL [SI: >10 mmol/L]): <25%.

Time below range (<70 mg/dL [SI: <3.9 mmol/L]): <4%.

Older adults (≥65 years of age) (ADA 2025):

Note: Consider less strict targets in patients who are using insulin and/or insulin secretagogues (sulfonylureas, meglitinides) (ES [LeRoith 2019]).

HbA_{1c}: <7% to 7.5% (healthy); <8% (complex/intermediate health). **Note:** Individualization may be appropriate based on patient and caregiver preferences and/or presence of cognitive impairment. In patients with very complex or poor health (ie, limited remaining life expectancy), consider making therapy decisions based on avoidance of hypoglycemia and symptomatic hyperglycemia rather than HbA_{1c} level.

Preprandial capillary blood glucose: 80 to 130 mg/dL (SI: 4.4 to 7.2 mmol/L) (healthy); 90 to 150 mg/dL (SI: 5 to 8.3 mmol/L) (complex/intermediate health); 100 to 180 mg/dL (SI: 5.6 to 10 mmol/L) (very complex/poor health).

Bedtime capillary blood glucose: 80 to 180 mg/dL (SI: 4.4 to 10 mmol/L) (healthy); 100 to 180 mg/dL (SI: 5.6 to 10 mmol/L) (complex/intermediate health); 110 to 200 mg/dL (SI: 6.1 to 11.1 mmol/L) (very complex/poor health).

Continuous glucose monitoring:

Time in range (70 to 180 mg/dL [SI: 3.9 to 10 mmol/L]): ~70% (healthy); ~50% (complex/intermediate health).

Time below range (<70 mg/dL [SI: <3.9 mmol/L]): <4% (healthy); <1% (complex/intermediate health).

Pregnant patients:

HbA_{1c}: Pregestational diabetes (type 1 or type 2) (ADA 2025):

Note: Due to physiological changes in RBC turnover, HbA_{1c} alone is not a sufficient measure of glycemic control during pregnancy; blood glucose monitoring is required.

Preconception (patients planning for pregnancy): <6.5%.

During pregnancy <6% (if can be achieved without significant hypoglycemia) or <7% if needed to prevent hypoglycemia.

Capillary blood glucose: **Note:** Less stringent targets may be appropriate if goals cannot be achieved without causing significant hypoglycemia. Lower limits of glucose ranges do not apply in patients with gestational diabetes mellitus who are not treated with insulin, or in patients with pregestational type 2 diabetes mellitus who are treated with nutrition alone (ADA 2025).

Fasting: 70 to 95 mg/dL (SI: 3.9 to 5.3 mmol/L).

Postprandial: 110 to 140 mg/dL (SI: 6.1 to 7.8 mmol/L) (at 1 hour) **or** 100 to 120 mg/dL (SI: 5.6 to 6.7 mmol/L) (at 2 hours).

Continuous glucose monitoring:

Note: Target ranges apply to all patients; however, time spent within each range are only established in patients with preexisting type 1 diabetes.

Time in range (63 to 140 mg/dL [SI: 3.5 to 7.8 mmol/L]): >70%.

Time above range (>140 mg/dL [SI: >7.8 mmol/L]): <25%.

Time below range (<63 mg/dL [SI: <3.5 mmol/L]): <4%.

Perioperative care (ADA 2025): Target glucose range during perioperative period (within 4 hours of surgery): 100 to 180 mg/dL (SI: 5.6 to 10 mmol/L).

Recommendations for hospitalized adult patients with hyperglycemia:

Critically ill patients: Target glucose range: 140 to 200 mg/dL (SI: 7.8 to 11.1 mmol/L) is appropriate for most patients; may consider a lower target range (eg, 110 to 140 mg/dL [SI: 6.1 to 7.8 mmol/L]) if it can be achieved with a low risk of hypoglycemia. Refer also to institutional protocols (SCCM [Honarmand 2024]).

Noncritically ill patients: Target glucose range: 100 to 180 mg/dL (SI: 5.6 to 10 mmol/L); less stringent targets (eg, ≤250 mg/dL [SI: ≤13.9 mmol/L]) may be appropriate in select patients (eg, limited life expectancy, advanced kidney failure, high hypoglycemia risk) to avoid hypoglycemia (ADA 2025).

Classification of hypoglycemia (ADA 2025):

Level 1: 54 to 70 mg/dL (SI: 3 to 3.9 mmol/L); hypoglycemia alert value; initiate fast-acting carbohydrate (eg, glucose) treatment.

Level 2: <54 mg/dL (SI: <3 mmol/L); threshold for neuroglycopenic symptoms; requires immediate action.

Level 3: Hypoglycemia associated with a severe event characterized by altered mental and/or physical status requiring assistance.

Mechanism of Action

Insulin acts via specific membrane-bound receptors on target tissues to regulate metabolism of carbohydrate, protein, and fats. Target organs for insulin include the liver, skeletal muscle, and adipose tissue.

Within the liver, insulin stimulates hepatic glycogen synthesis. Insulin promotes hepatic synthesis of fatty acids, which are released into the circulation as lipoproteins. Skeletal muscle effects of insulin include increased protein synthesis and increased glycogen synthesis. Insulin stimulates lipoprotein lipase synthesis and activity; this results in hydrolysis of triglycerides into free fatty acids and storage

of free fatty acids in adipocytes, thereby reducing circulating triglyceride levels (Rawla 2018; Sadur 1982; Song 2019). In addition, insulin stimulates the cellular uptake of amino acids and increases cellular permeability to several ions, including potassium, magnesium, and phosphate. By activating sodium-potassium ATPases, insulin promotes the intracellular movement of potassium.

Normally secreted by the pancreas, insulin products are manufactured for pharmacologic use through recombinant DNA technology using either *E. coli* or *Saccharomyces cerevisiae*. Regular insulin has an identical structure to that of native human insulin. Insulins are categorized based on the onset, peak, and duration of effect (eg, rapid-, short-, intermediate-, and long-acting insulin). Insulin regular is a short-acting insulin analog.

The benefit of high-dose insulin in patients experiencing toxicity secondary to a calcium channel blocker (CCB) or beta blocker is not fully understood. It is hypothesized insulin improves myocyte glucose uptake and utilization; insulin may also act directly as a concentration-dependent inotrope. Exogenous insulin administration may help to overcome the hypoinsulinemia caused by CCB-induced blockage of L-type calcium channels in pancreatic beta cells (Krenz 2018).

Pharmacokinetics (Adult Data Unless Noted)

Note: Onset and duration of hypoglycemic effects depend upon the route of administration (absorption and onset of action are more rapid after deeper IM injections than after SUBQ), site of injection (onset and duration are progressively slower with SUBQ injection into the abdomen, arm, buttock, or thigh respectively), volume and concentration of injection, and the preparation administered. Rate of absorption, onset, and duration of activity may be affected by exercise, presence of lipodystrophy, local blood supply, and/or temperature. Insulin regular is available in a U-100 (100 units/mL) formulation and a concentrated U-500 (500 units/mL) formulation, which have different pharmacodynamic/kinetic profiles.

Onset of action:

U-100 formulation: IV: ~10 to 21 minutes; SUBQ: ~30 minutes.

U-500 formulation: SUBQ: <15 minutes.

Peak effect:

U-100 formulation: IV infusion: ~5 hours; SUBQ: 1.5 to 3.5 hours.

U-500 formulation: SUBQ: 4 to 8 hours (dose dependent).

Duration:

U-100 formulation: IV infusion: 1.5 hours after stopping infusion; SUBQ: ~8 hours.

U-500 formulation: SUBQ: 13 to 24 hours.

Distribution: V_d : **U-100 formulation:** IV: 0.32 to 0.67 L/kg.

Bioavailability: **U-100 formulation:** SUBQ: 48% to 89%.

Half-life elimination:

U-100 formulation: IV: ~0.3 to 1 hour (dose dependent); SUBQ: 1.5 hours.

U-500 formulation: SUBQ: 4.5 hours.

Time to peak, plasma:

U-100 formulation: SUBQ: 0.5 to 2.5 hours.

U-500 formulation: SUBQ: 0.5 to 8 hours.

Patient Counseling Points

(For additional information see "Insulin regular: Patient drug information")

What is this drug used for?

- It is used to lower blood sugar in patients with high blood sugar (diabetes).

All drugs may cause side effects. However, many people have no side effects or only have minor side effects. Call your doctor or get medical help if any of these side effects or any other side effects bother you or do not go away:

- Weight gain
- Irritation where the shot is given

WARNING/CAUTION: Even though it may be rare, some people may have very bad and sometimes deadly side effects when taking a drug. Tell your doctor or get medical help right away if you have any of the following signs or symptoms that may be related to a very bad side effect:

- Low potassium levels like muscle pain or weakness, muscle cramps, or a heartbeat that does not feel normal
- Swelling in the arms or legs
- Thick skin, pits, or lumps where the injection was given
- Low blood sugar like dizziness or passing out, blurred eyesight, mood changes, slurred speech, headache, feeling sleepy or weak, shaking, fast heartbeat, confusion, hunger, sweating, or seizures
- Signs of an allergic reaction, like rash; hives; itching; red, swollen, blistered, or peeling skin with or without fever; wheezing; tightness in the chest or throat; trouble breathing, swallowing, or talking; unusual hoarseness; or swelling of the mouth, face, lips, tongue, or throat.

Note: This is not a comprehensive list of all side effects. Talk to your doctor if you have questions.

Consumer Information Use and Disclaimer: This information should not be used to decide whether or not to take this medicine or any other medicine. Only the healthcare provider has the knowledge and training to decide which medicines are right for a specific patient. This information does not endorse any medicine as safe, effective, or approved for treating any patient or health condition. This is only a limited summary of general information about the medicine's uses from the patient education leaflet and is not intended to be comprehensive. This limited summary does NOT include all information available about the possible uses, directions, warnings, precautions, interactions, adverse effects, or risks that may apply to this medicine. This information is not intended to provide medical advice, diagnosis or treatment and does not replace information you receive from the healthcare provider. For a more detailed summary of information about the risks and benefits of using this medicine, please speak with your healthcare provider and review the entire patient education leaflet.

Brand Names: International

International Brand Names by Country

For country code abbreviations (show table)

- (AE) United Arab Emirates: Actrapid hm | Humulin regular | Jusline R;
- (AR) Argentina: Actrapid hm | Densulin regular | Humulin r | Insulina Densulin R | Insulina Humulin R | Insuman r | Novolin r;
- (AT) Austria: Actrapid | Huminsulin normal | Insulin hm actrapid | Insuman rapid;
- (AU) Australia: Actrapid | Actrapid hm | Humulin r;
- (BD) Bangladesh: Actrapid flexpen | Ansulín R | Diasulin n | Diasulin r | Humulin r | Insulet n | Insulet r | Insulin actrapid hm | Insuman rapid | Mysulin r;
- (BE) Belgium: Actrapid | Humuline regular | Insuman rapid;
- (BF) Burkina Faso: Insulet rapide | Insuman rapid;
- (BG) Bulgaria: Actrapid hm | Actrapid hm novolet | Actrapid novolet | Humulin r | Insuman rapid;
- (BR) Brazil: Bahiafarma insulina humana r | Biohulin regular | Humulin r | Insuliv r | Insuman r | Ivb insulina humana r | Novolin r | Wosulin r;
- (CH) Switzerland: Actrapid hm | Actrapid novolet;

- (CI) Côte d'Ivoire: Insuman rapid;
- (CL) Chile: Actrapid hm | Humulin r | Insuman;
- (CN) China: Actrapid | Biosynthetic Human Insulin | Gan shu lin r | Humulin r | Novolin r;
- (CO) Colombia: Actrapid hm | Gansulin r | Humulin c | Humulin r | Insugen r | Insulex r;
- (CR) Costa Rica: Novolin r;
- (CZ) Czech Republic: Actrapid hm | Actrapid innolet | Actrapid novolet | Humulin r | Insulin-HM R | Insuman rapid | Velosulin HM;
- (DE) Germany: Actrapid | Actrapid flexpen | Actrapid novolet | Berlinsulin h normal | Huma ject normal | Huminsulin normal | Humulin normal | Insulin B. Braun ratiopharm rapid | Insuman rapid;
- (DO) Dominican Republic: Humulin regular | Insuman r | Novolin r | Recomulin r;
- (EC) Ecuador: Actrapid | Humulin r | Insuman r | Novolin r;
- (EE) Estonia: Actrapid | Actrapid hm | Humaject R | Humulin r | Insuman rapid | Isuhuman rapid;
- (EG) Egypt: Actrapid hm | Humulin regular | Insulin h r | Insuman rapid;
- (ES) Spain: Actrapid | Actrapid innolet | Actrapid novolet | Humaplast regular | Humulina regular;
- (ET) Ethiopia: Actrapid | Biosynthetic Human Insulin | Jusline R;
- (FI) Finland: Actrapid | Humulin regular | Insuman rapid;
- (FR) France: Actrapid hm | Actrapid novolet | Humulin regular | Insuman rapid | Umuline rapide | Velosuline hm;
- (GB) United Kingdom: Actrapid | Actrapid novolet | Humulin s | Insuman rapid | Pur in Neutral | Velosulin;
- (GR) Greece: Actrapid | Actrapid novolet | Humulin regular | Insulin actrapid hm | Insuman rapid | Velosulin HM;
- (HK) Hong Kong: Actrapid hm | Humulin r;
- (HR) Croatia: Actrapid innolet;
- (HU) Hungary: Actrapid | Actrapid novolet | Humulin r | Ins actrapid hm | Insulin actrapid hm(ge) | Insuman rapid;
- (ID) Indonesia: Actrapid hm | Actrapid novolet | Humulin r | Insuman rapid;
- (IE) Ireland: Actrapid | Humulin r | Humulin s | Insuman;
- (IL) Israel: Actrapid | Humulin r;
- (IN) India: Actrapid flexpen | Actrapid hm | Actrapid novolet | Humaject R | Human actrapid | Human Insunorm | Human monosulin | Human rapidica | Human-fastact | Humarap | Huminsulin-r | Insugen r | Insuman rapid | Recosulin-r | Wosulin r;
- (IT) Italy: Actrapid | Bio insulin r | Humulin r | Insuman rapid;
- (JO) Jordan: Actrapid hm | Gensulin r | Humulin r;
- (JP) Japan: Actrapid human | Humacart r | Humulin r | Innolet r | Novolin r | Novolin r flexpen | Velosulin;
- (KE) Kenya: Actrapid | Humodar c25 | Humulin r | Jusline R;
- (KR) Korea, Republic of: Actrapid hm | Humaject R | Humulin r | Novolet r | Novolin r | Velosulin Human;
- (KW) Kuwait: Actrapid hm | Actrapid novolet | Humulin regular;
- (LB) Lebanon: Actrapid | Humulin regular | Insuman rapid | Velosulin HM;
- (LT) Lithuania: Actrapid hm | Biogulin | Humulin r | Insuman rapid;
- (LU) Luxembourg: Actrapid | Humaject Regular | Humuline regular | Velosuline hm;
- (LV) Latvia: Actrapid hm | Biogulin | Humulin r | Insuman rapid;
- (MA) Morocco: Insulet r | Insulet rapide | Umuline rapide;
- (MX) Mexico: Humulin r | Insulex r | Insuman r | Novolin r | Wozulim r;
- (MY) Malaysia: Actrapid flexpen | Actrapid hm | Actrapid novolet | Humulin r;
- (NL) Netherlands: Actrapid | Actrapid flexpen | Actrapid novolet | Humuline regular | Inpremia | Insuman rapid | Velosulin;
- (NO) Norway: Actrapid | H-tronin | Humulin r | Humulin regular | Insuman rapid;
- (NZ) New Zealand: Actrapid | Humulin r;
- (PA) Panama: Novolin r;
- (PE) Peru: Actrapid hm | Humulin c | Humulin r | Insuman r;
- (PH) Philippines: Actrapid hm | Actrapid novolet | Glysolin r | Humulin r | Insulyf r | Insuman rapid | Jusline R | Wosulin r;

- (PK) Pakistan: Actrapid hm | Dongsulin R | Humulin regular | Innogen R | Insuget R | Insuman rapid | Zansulin regular;
- (PL) Poland: Actrapid hm novolet | Gensulin r | Humulin r | Insuman rapid;
- (PR) Puerto Rico: Humulin r | Myxredlin | Novolin r | Novolin r innolet | Relion/Novolin R | Velosulin rdna br;
- (PT) Portugal: Actrapid | Humulin regular | Insulina actrapid hm | Insuman rapid;
- (PY) Paraguay: Actrapid hm | Humulin r | Insubest r | Insuman r | Sulina r;
- (QA) Qatar: Actrapid | Actrapid Penfill | Humulin R | Insuman Rapid;
- (RO) Romania: Humulin r | Insuman rapid;
- (RU) Russian Federation: Actrapid hm | Berlinsulin h normal | Biogulin | Biosulin r | Gensulin r | Humodar r | Humulin r | Humulin regular | Ins actrapid hm | Ins actrapid nm | Insuman rapid | Insuman rapid gt | Rosinsulin p | Wozulim r;
- (SA) Saudi Arabia: Actrapid | Actrapid hm | Humulin r | Humulin regular;
- (SE) Sweden: Actrapid | Actrapid novolet | Insuman rapid | Insuman Rapid SoloStar;
- (SG) Singapore: Actrapid hm | Actrapid novolet | Humulin r;
- (SI) Slovenia: Actrapid hm | Humulin r;
- (SK) Slovakia: Actrapid hm | Humulin r | Insulin-HM R | Insuman rapid;
- (SR) Suriname: Humulin r | Insugen r;
- (SV) El Salvador: Insulex r;
- (TH) Thailand: Actrapid hm | Biosulin r | Gensulin r | Humulin r | Winsulin r;
- (TN) Tunisia: Actrapid | Insuline actrapid hm | Insuman rapid | Jusline R;
- (TR) Turkey: Actrapid hm | Actrapid novolet | Humulin regular | Insulin Protaphan HM;
- (TW) Taiwan: Actrapid hm | Humulin r | Insulin Regular hm | Velosulin | Velosulin HM;
- (UA) Ukraine: Actrapid flexpen | Actrapid hm | Gensulin r | Humulin regular | Insular active | Insuman rapid;
- (UG) Uganda: Actrapid | Actrapid flexpen | Biosulin r | Gensulin r | Humulin regular | Insugen r | Insuman rapid | Wosulin r;
- (UY) Uruguay: Actrapid hm | Humulin r | Insubest r | Insuman r | Wosulin r;
- (VE) Venezuela, Bolivarian Republic of: Humulin c | Humulin r | Insulex r | Insuman r;
- (ZA) South Africa: Actrapid | Biosulin r | Humaject R | Humulin r

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